

NCHC

Physics-Informed Neural Networks (PINNs): A Novel Computing Paradigm for AI-driven Engineering & Scientific Simulations

National Center for High-performance Computing
National Institutes of Applied Research

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Nomenclature and Acronym

Physics-Informed Neural Networks (PINNs)

Physics-Informed Neural Network (PINN)

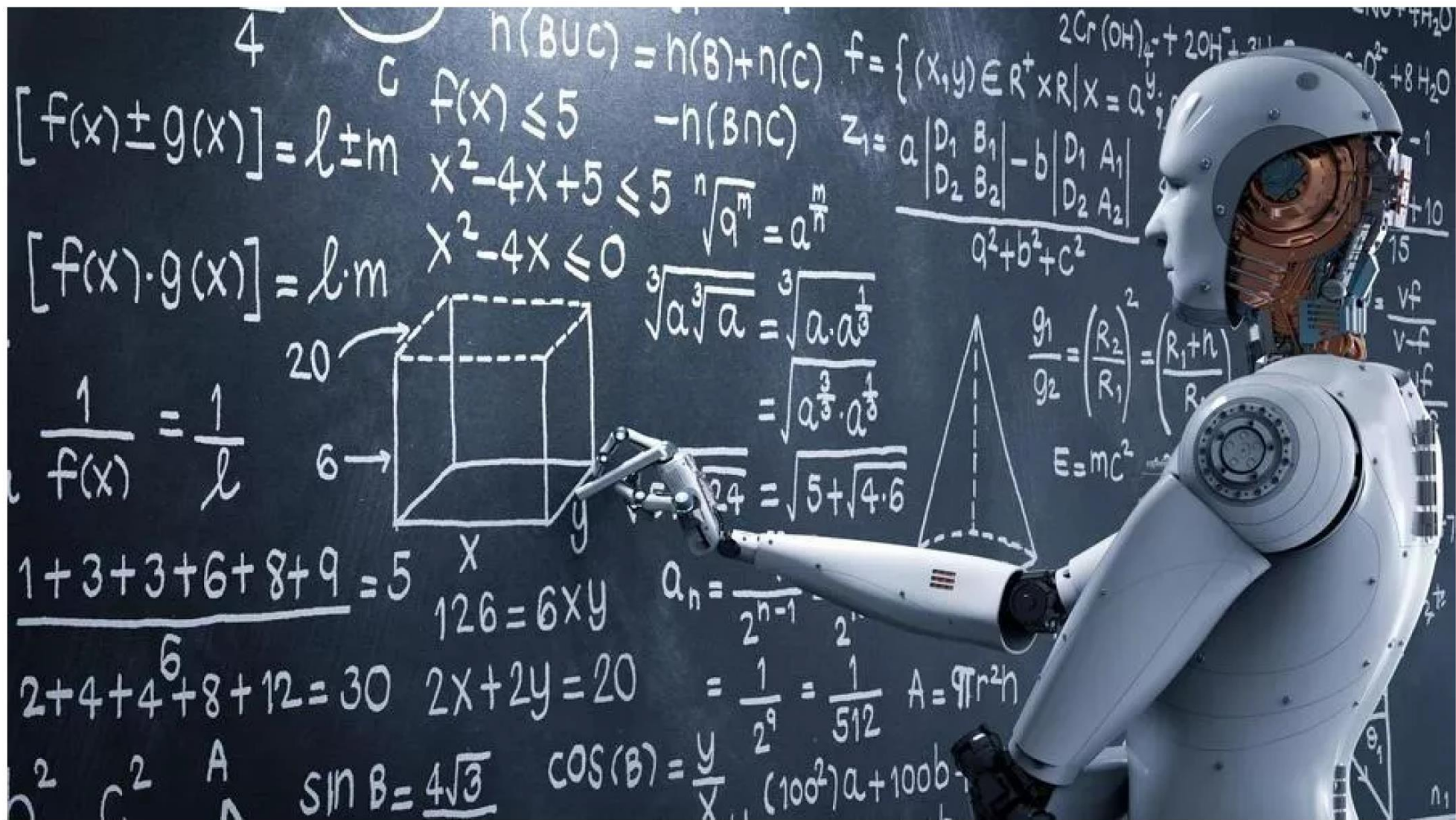
Physics-Informed Adversarial Deep Learning (PADL)

Physics-Informed Gradient Descent Learning (PGDL)

Knowledge-Informed Machine Learning (KIML)

Scientific Machine Learning (SciML)

...

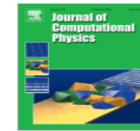


From Physics-Based to Machine Learning Approaches - “PINNs”

Raissi et al. (2017) introduced the **Physics-Informed Neural Networks** (PINNs), which can integrate physics information to train neural networks.



Journal of Computational Physics
Volume 378, 1 February 2019, Pages 686-707



Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations

M. Raissi ^a, P. Perdikaris ^b  , G.E. Karniadakis ^a

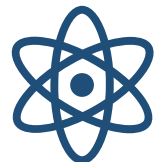
Citations: 22,740+

^a Division of Applied Mathematics, Brown University, Providence, RI, 02912, USA

^b Department of Mechanical Engineering and Applied Mechanics, University of Pennsylvania, Philadelphia, PA, 19104, USA

arXiv:1711.10561; arXiv:1711.10566 (2017)

Physics



Numerical
Solver



PINNs



Data
Driven

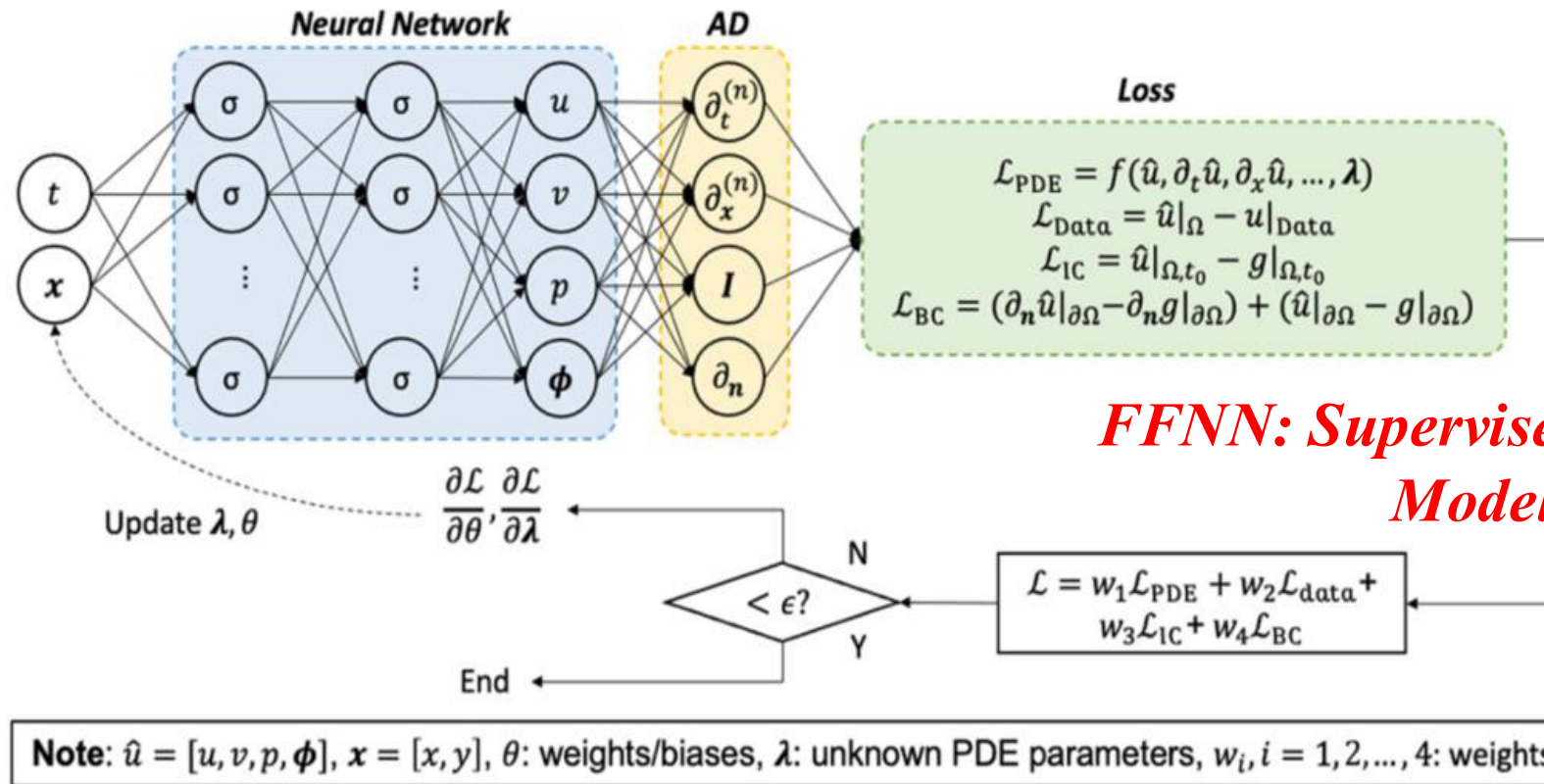
AI



- Strengths to solve high-dimensional and inverse problems
- Source Codes uploaded to GitHub for free downloads - <https://github.com/maziarraissi/PINNs>

How Does PINNs Work?

Idea: Embed PDEs, BCs/ or ICs & Data into Loss Functions by Automatic Differentiation (AD)



Characteristics:

- Particle-based or Meshless/Meshfree method
- Seamlessly integrate data and physics
- A unified, flexible framework (Function Approximator $\leftrightarrow$ Numerical Solver)

Ongoing Collaborative Projects

■ Prof. Yih-Chin Tai, Dept. of Hyd. and Ocean Eng., NCKU

- Adaptive Neural Particle Method in Modelling Fluid Flows
- Integration of PINNs Simulation Results with Omniverse Visualization Platform
- **Adaptive refinement Neural Particle Method (arNPM) for Granular Flows**

■ Prof. Ming-Jyh Chern, Dean of College of Engineering, NTUST

- Assessing PINNs Performance with sparse noisy velocity data
- Numerical Error Estimation with Physics Informed Neural Networks
- **Using Physics-Informed Neural Networks (PINNs) to Transform Turbulence Models**

■ Prof. Chung-Gang Lee, Dept. of Mech. Eng., NCKU

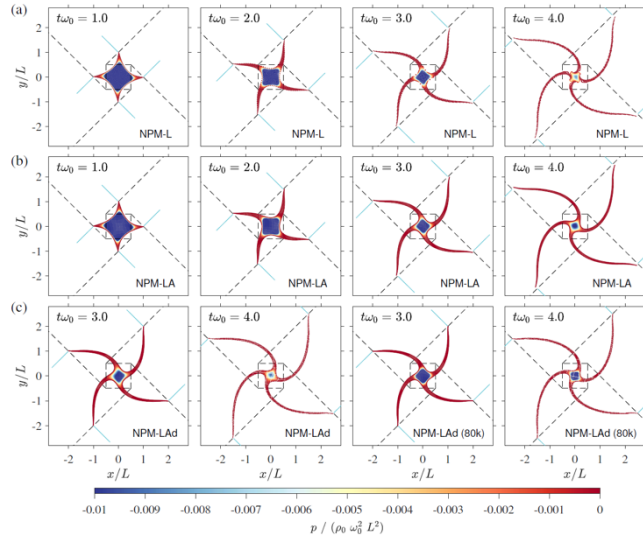
- Physics-Informed Neural Networks (PINNs) for heat transfer and flow interaction
- **Industry-Academia Cooperative Research Project from ASE Global**

PINNs related Journal Papers

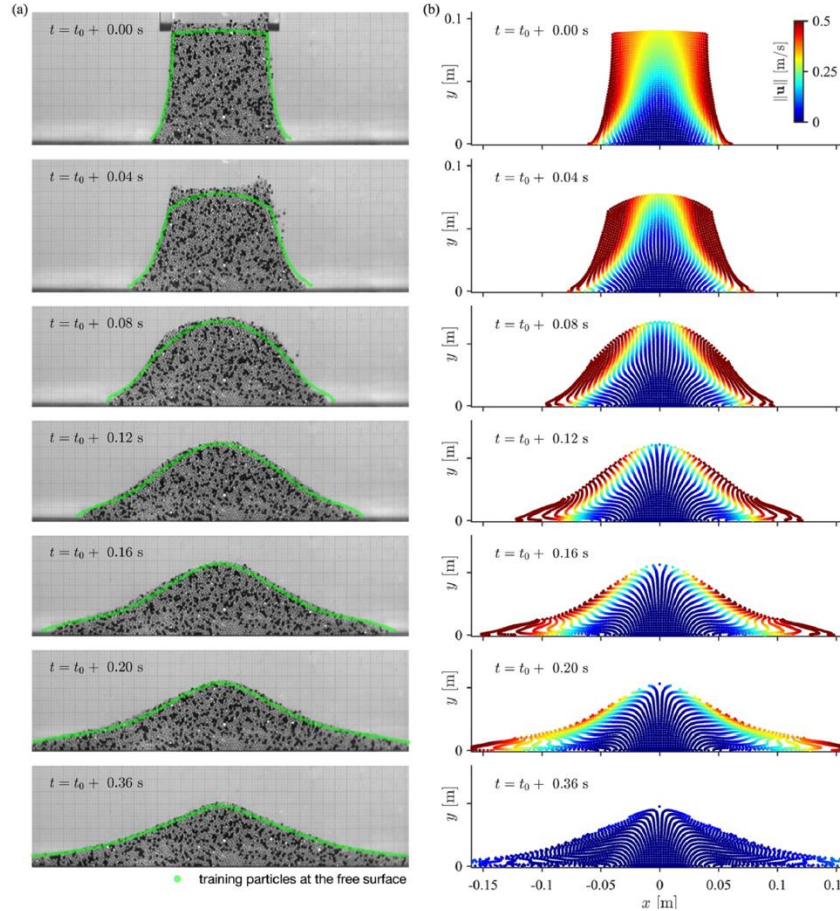
1. Edward Oliver Teng, Chung-Gang Lee*, and **Heng-Chuan Kan**. “Thermal Analysis and Flow Regime Characterization 1 of Fin-Embedded Heat Sinks Using Hybrid Physics-Informed Neural Networks”. *Physics of Fluids (I.F.=4.6)*, 2026. (under review)
2. Adhika Satyadharma, Ming-Jyh Chern, **Heng-Chuan Kan***, Jing-Yuan Fan, and Chun-Ying Yu. “Evaluating External Flow Simulations Domain Size Effects and Its Quantification Using Physics Informed Neural Network”. *Computers and Math. with Applications (I.F.=2.8)*, 2026. (under review)
3. Pei-Hsin Pai, Luca Sarno*, Yih-Chin Tai, and **Heng-Chuan Kan**. “An adaptive refinement Neural Particle Method for granular flows”. *Physics of Fluids (I.F.=4.6)*, 2025. (<https://doi.org/10.1063/5.0276235>)
4. Adhika Satyadharma, **Heng-Chuan Kan***, Ming-Jyh Chern, and Chun-Ying Yu. “Numerical Error Estimation with Physics Informed Neural Network”. *Computers & Fluids (I.F.=3.0)*, 2025. (doi:10.1016/j.compfluid.2025.106700)
5. Pei-Hsin Pai, **Heng-Chuan Kan***, Hock-Kiet Wong, and Yih-Chin Tai*. “A neural particle method with interface tracking and adaptive particle refinement for free surface flows”. *Communications in Computational Physics (I.F.=3.7)*, 36, pp. 1021-1052, 2024. (doi:10.4208/cicp.OA-2023-0235)
6. Adhika Satyadharma, Ming-Jyh Chern, **Heng-Chuan Kan***, Harinaldi, and James Julian. “Assessing Physics Informed Neural Network Performance with Sparse Noisy Velocity Data”. *Physics of Fluids (I.F.=4.6)*, 36, 103619, 2024. (doi:10.1063/5.0213522)

PINNs for Engineering & Scientific Simulations and Applications

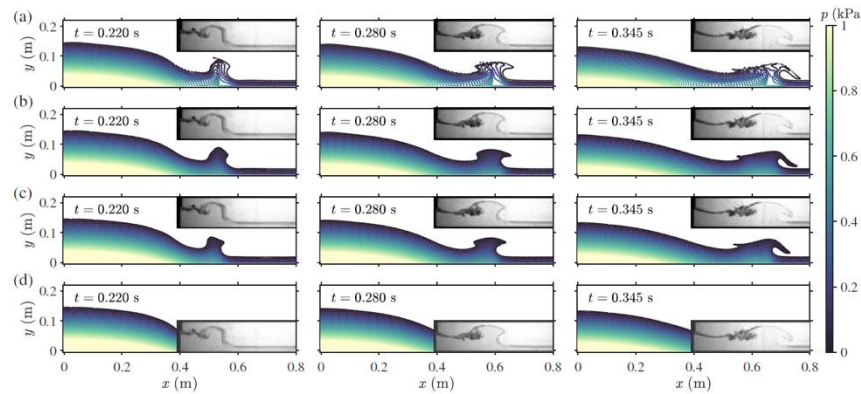
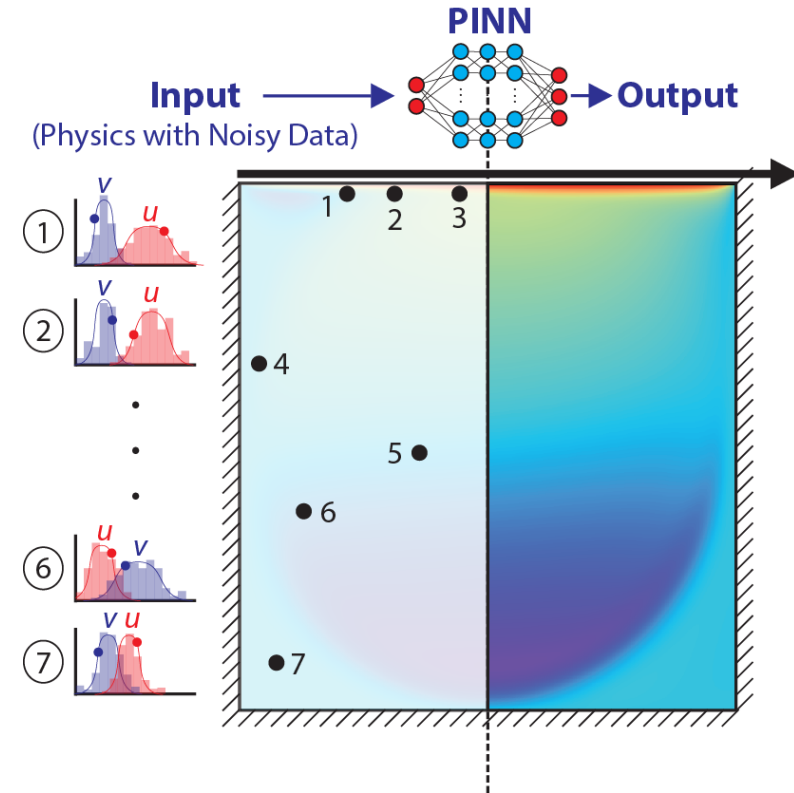
Adaptive Neural Particle Method for Free-Surface Flows



Non-Newtonian Fluid Modeling for Granular Column Collapsing



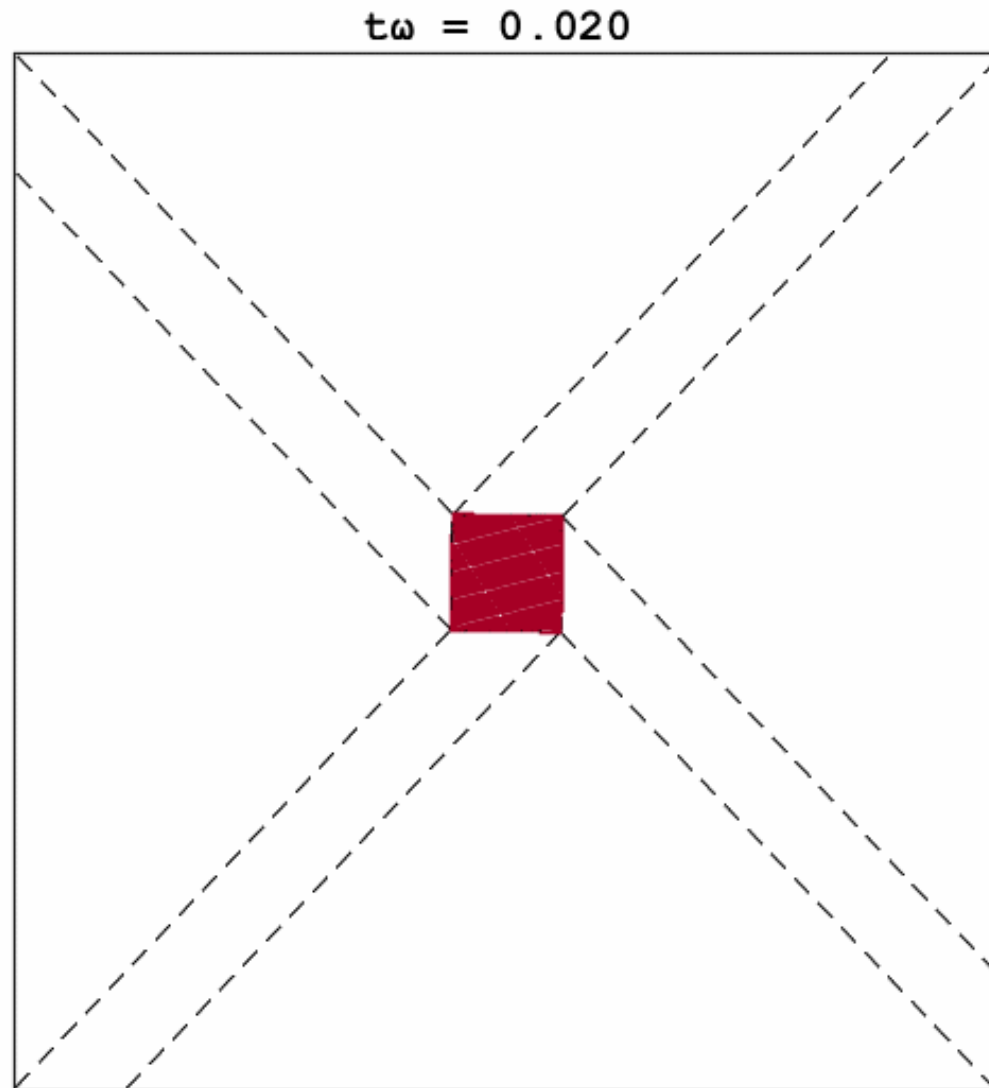
Enhancing PINNs Performance with Sparse Noisy Data



Pei-Hsin Pai*, Luca Sarno, Yih-Chin Tai, and Heng-Chuan Kan. “An adaptive refinement Neural Particle Method for granular flows”. Physics of Fluids (I.F.=4.6), 2025. (Accepted)

Adhika Satyadharma, Ming-Jyh Chern, Heng-Chuan Kan*, Harinaldi, and James Julian. “Assessing Physics Informed Neural Network Performance with Sparse Noisy Velocity Data”. Physics of Fluids, 36, 103619, 2024. (doi:10.1063/1.514352)

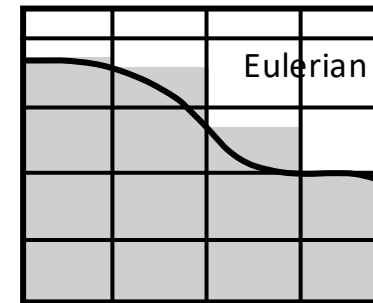
Motivation: Rotating Square Fluid Patch



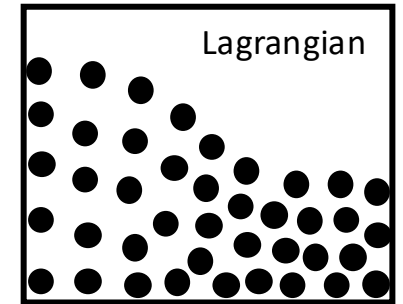
Challenges :

- Free-moving boundary

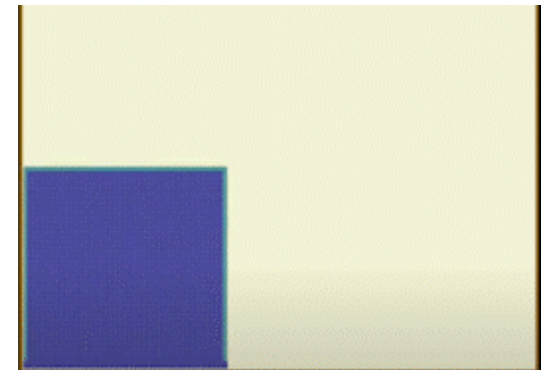
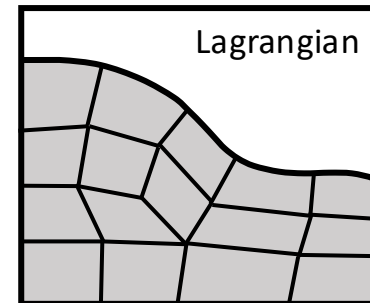
Fixed Mesh



Moving Particles

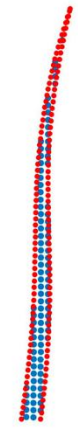
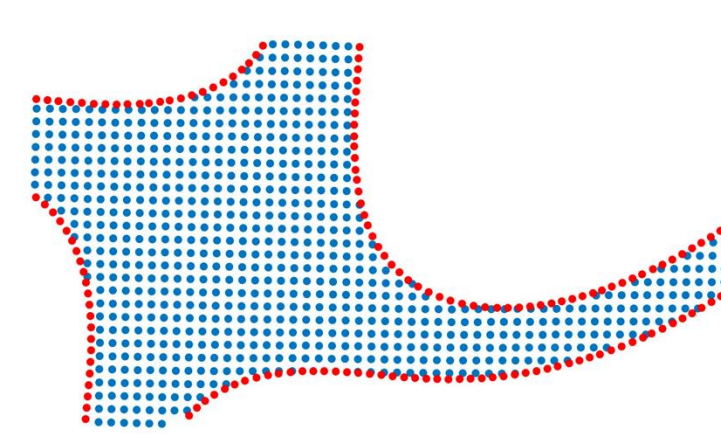
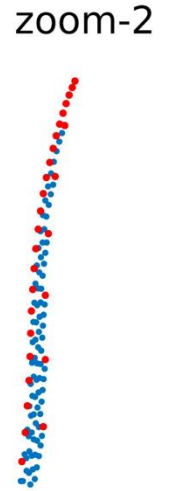
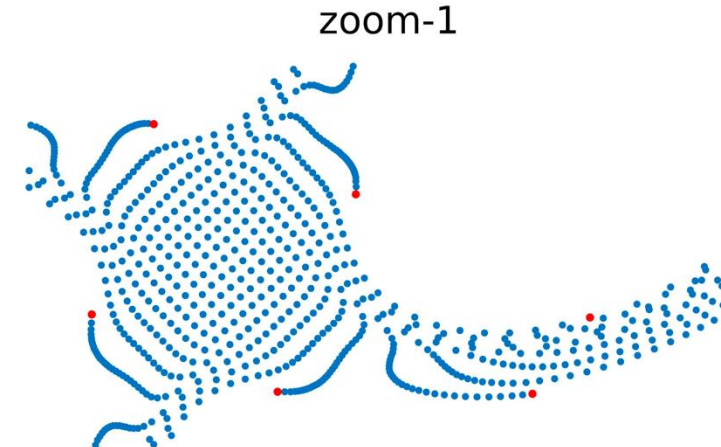
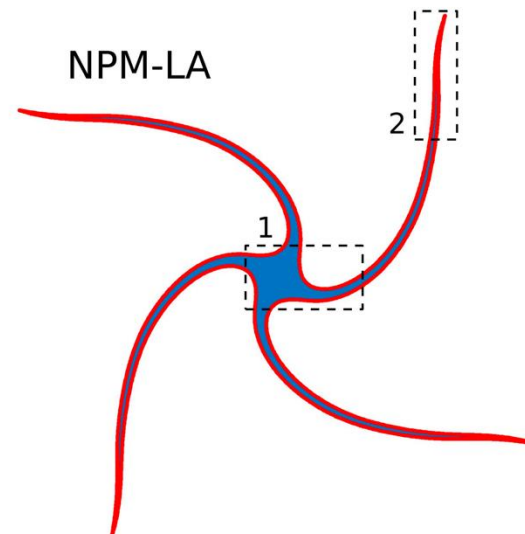
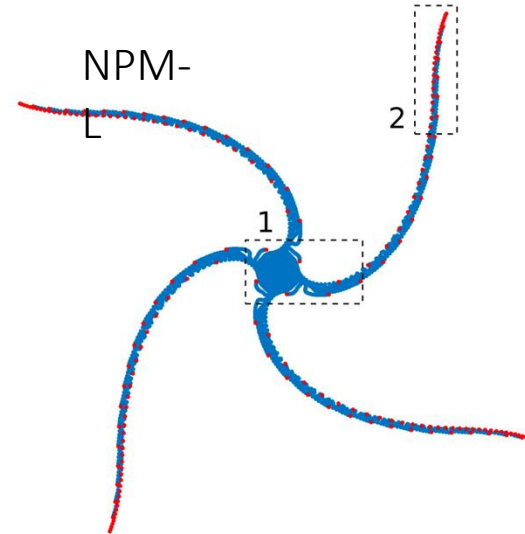
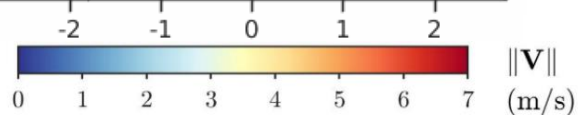
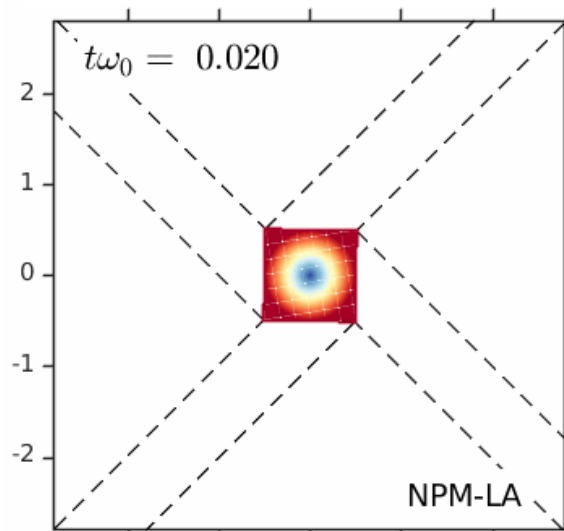
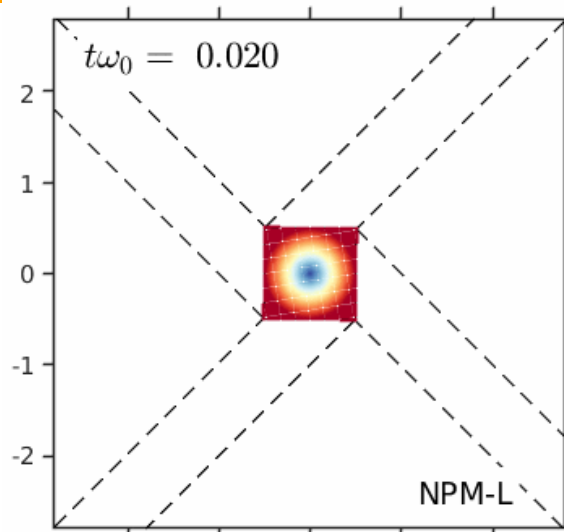


Moving Mesh



<https://youtu.be/XOxVki6l2Dc>

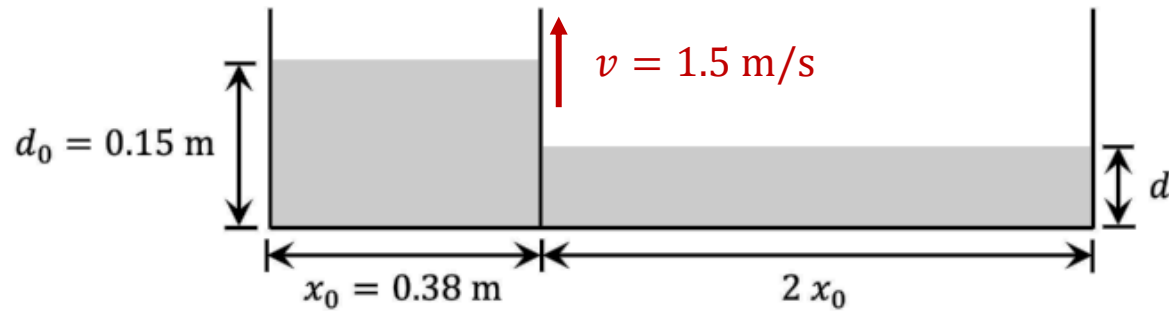
Rotating Square Fluid Patch Simulations



• interior particles • boundary particles

2-D Dam Break Simulations with Dry Bed and Wet Bed

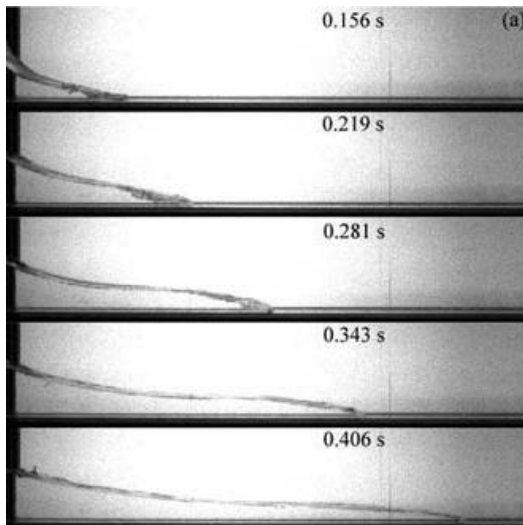
Experiment of János et al. (2004)



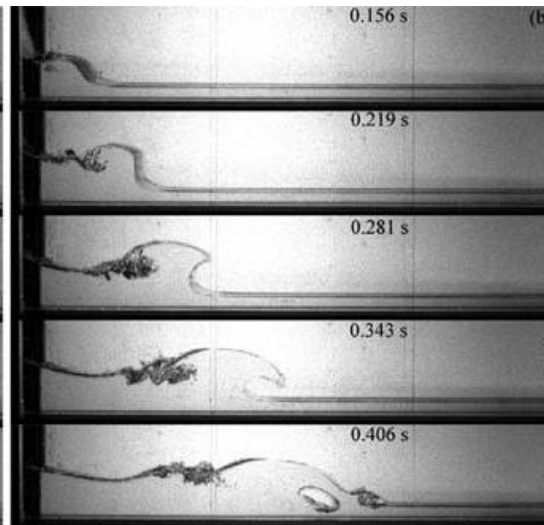
Info

$\rho = 1000 \text{ kg/m}^3$ Dry bed :
 $\mu = 0.001 \text{ Pa} \cdot \text{s}$ $dx = dy = 0.007 \text{ m}$ (1,347 pts)
 $\Delta t = 0.005 \text{ s}$ Wet bed :
 $dx = dy \approx 0.0034 \text{ m}$ (6,349 pts)

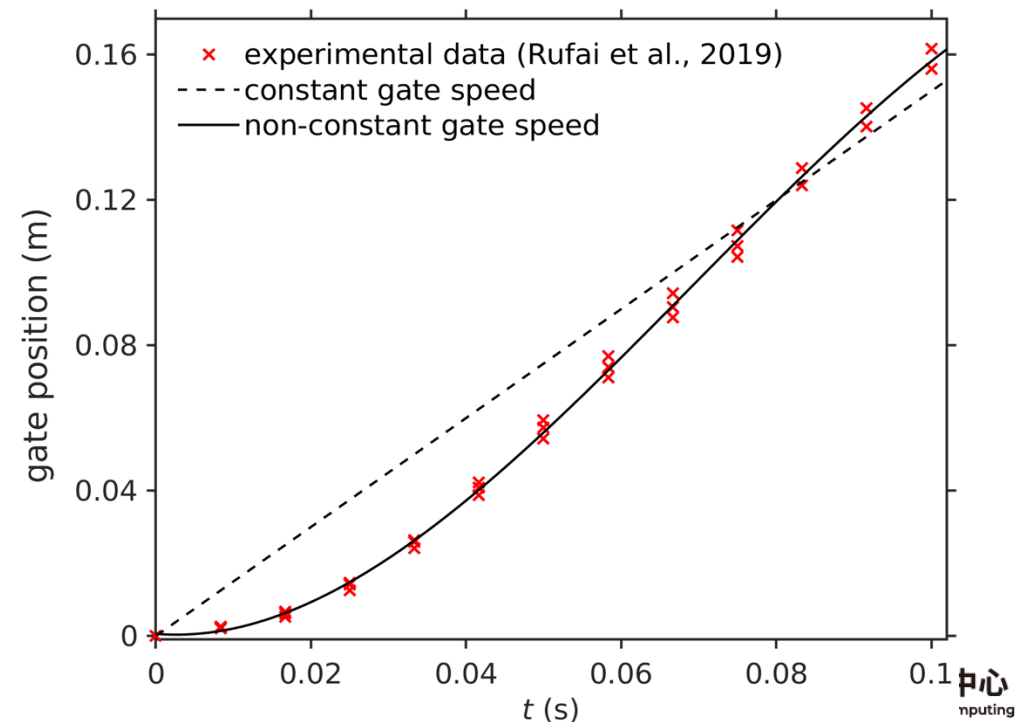
Dry Bed
($d = 0 \text{ m}$)



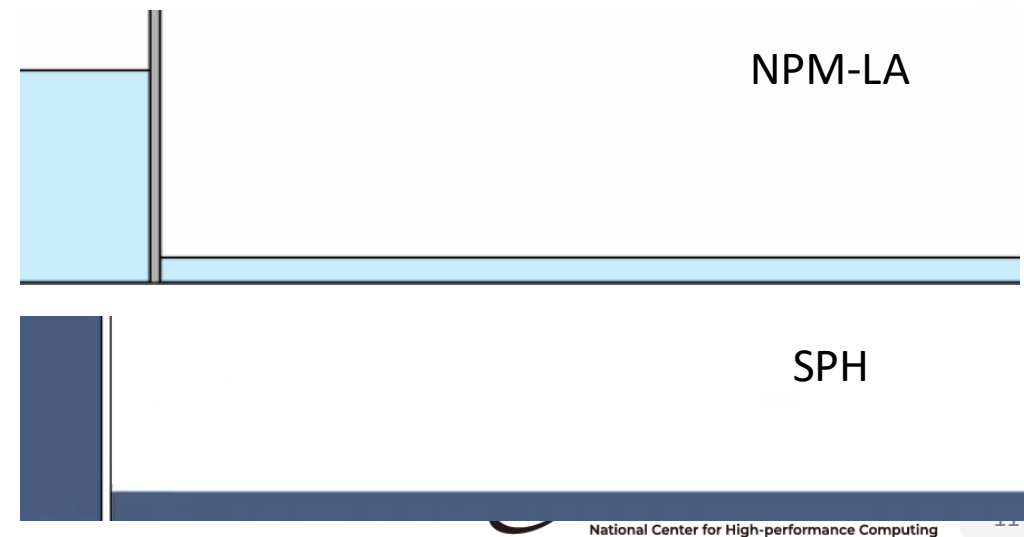
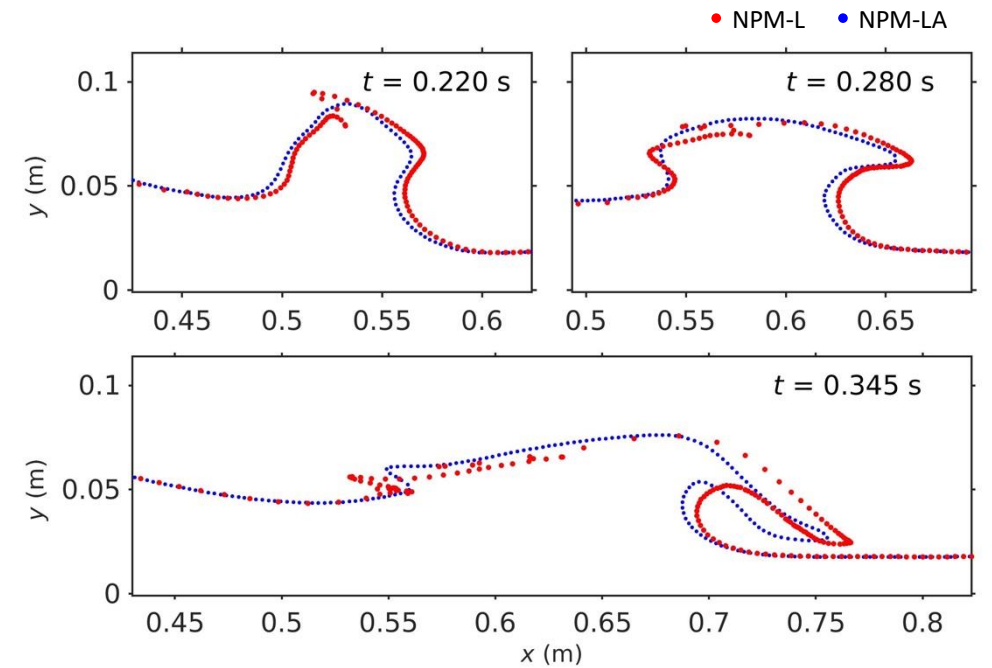
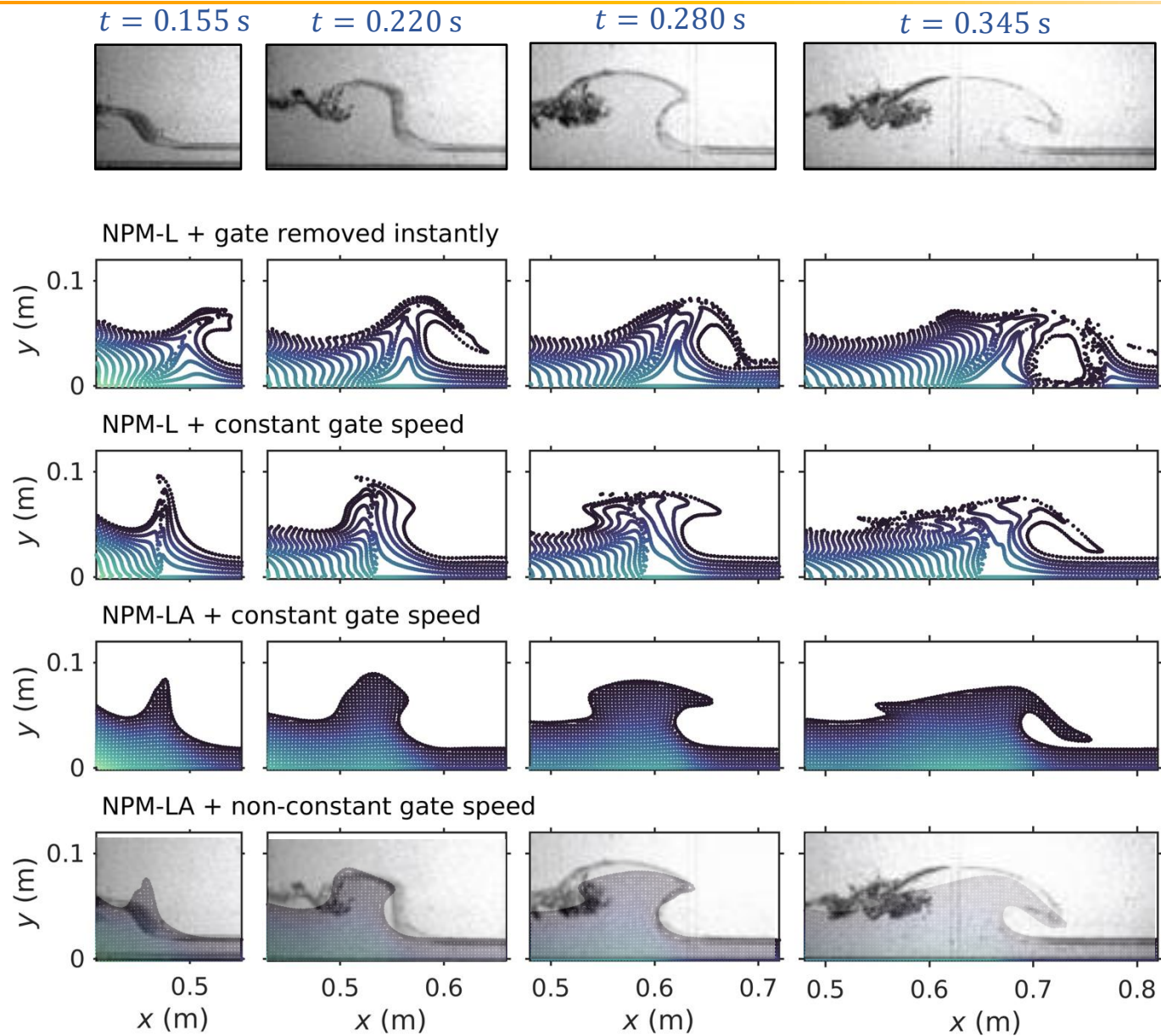
Wet Bed
($d = 0.018 \text{ m}$)



Constant/Non-Constant Gate Speed



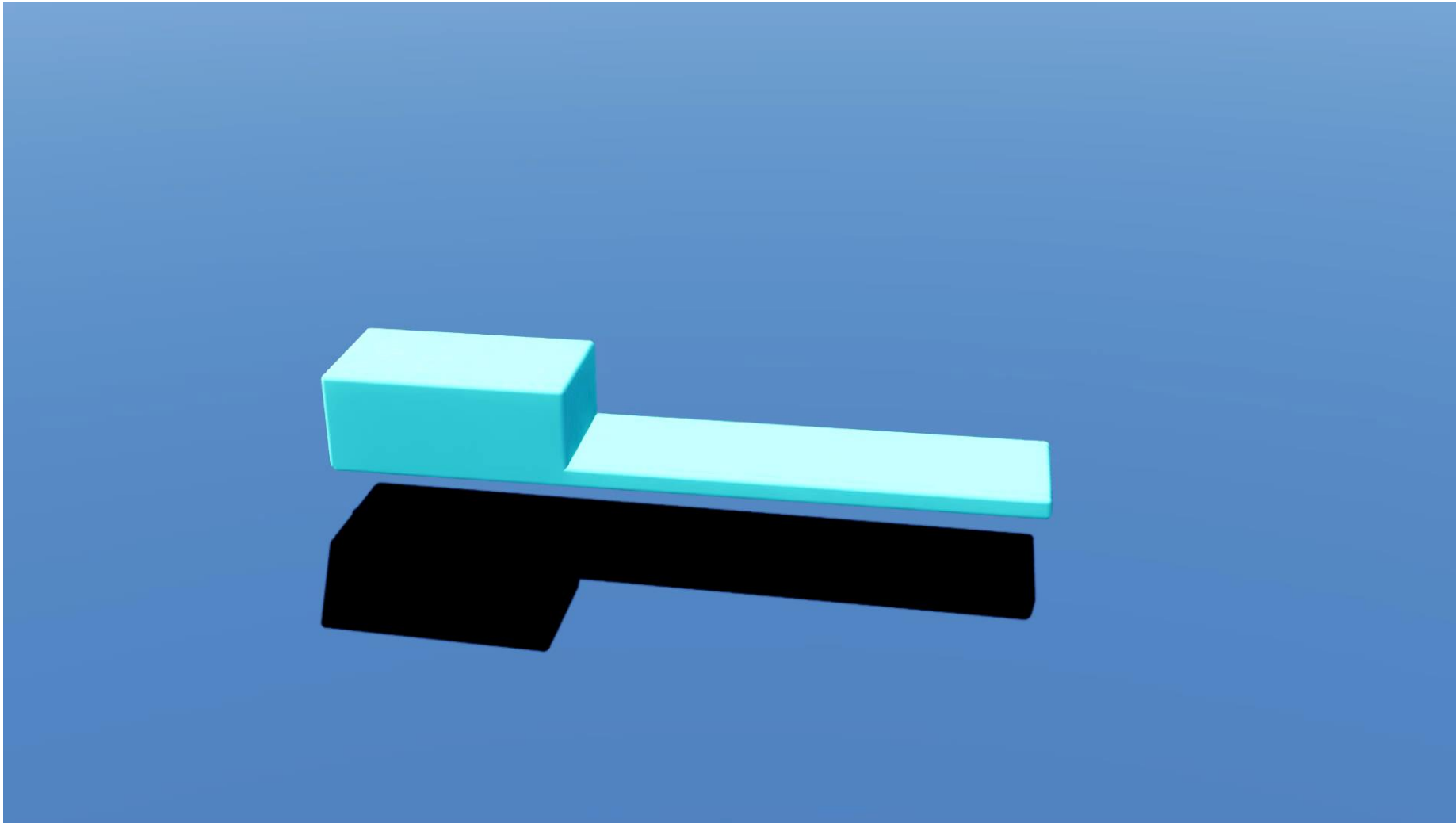
2-D Dam Break Simulations – Wet Bed Case



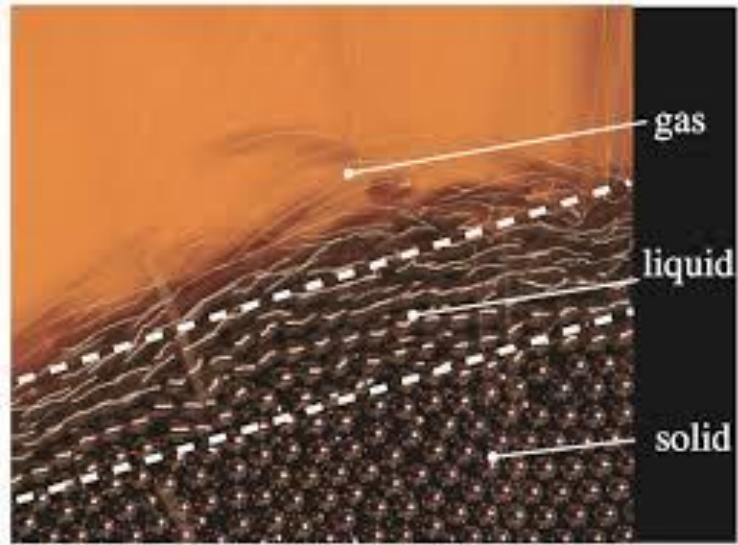
NVIDIA Modulus (PINNs) + Visualization: Omniverse Platform



NVIDIA Modulus (PINNs) + Visualization: Omniverse Platform



Our Target: Granular Flows



(Forterre et al., 2008)



<https://news.mit.edu/2018>



<https://www.swissinfo.ch>

$\mu(I)$ Rheology for dense granular flows (e.g., Jop et al. 2005, *JFM*; Jop et al. 2006, *Nature*)

$$\nabla \cdot \mathbf{u} = 0$$

$$\mathbf{a} = \frac{d\mathbf{u}}{dt} = \mathbf{g} - \frac{1}{\rho} \nabla p + \frac{1}{\rho} \nabla \cdot \mathbf{T}$$

$$\mathbf{T} = 2\eta \mathbf{D} = \frac{\sqrt{2}\mu(I)p}{\|\mathbf{D}\|} \mathbf{D}$$

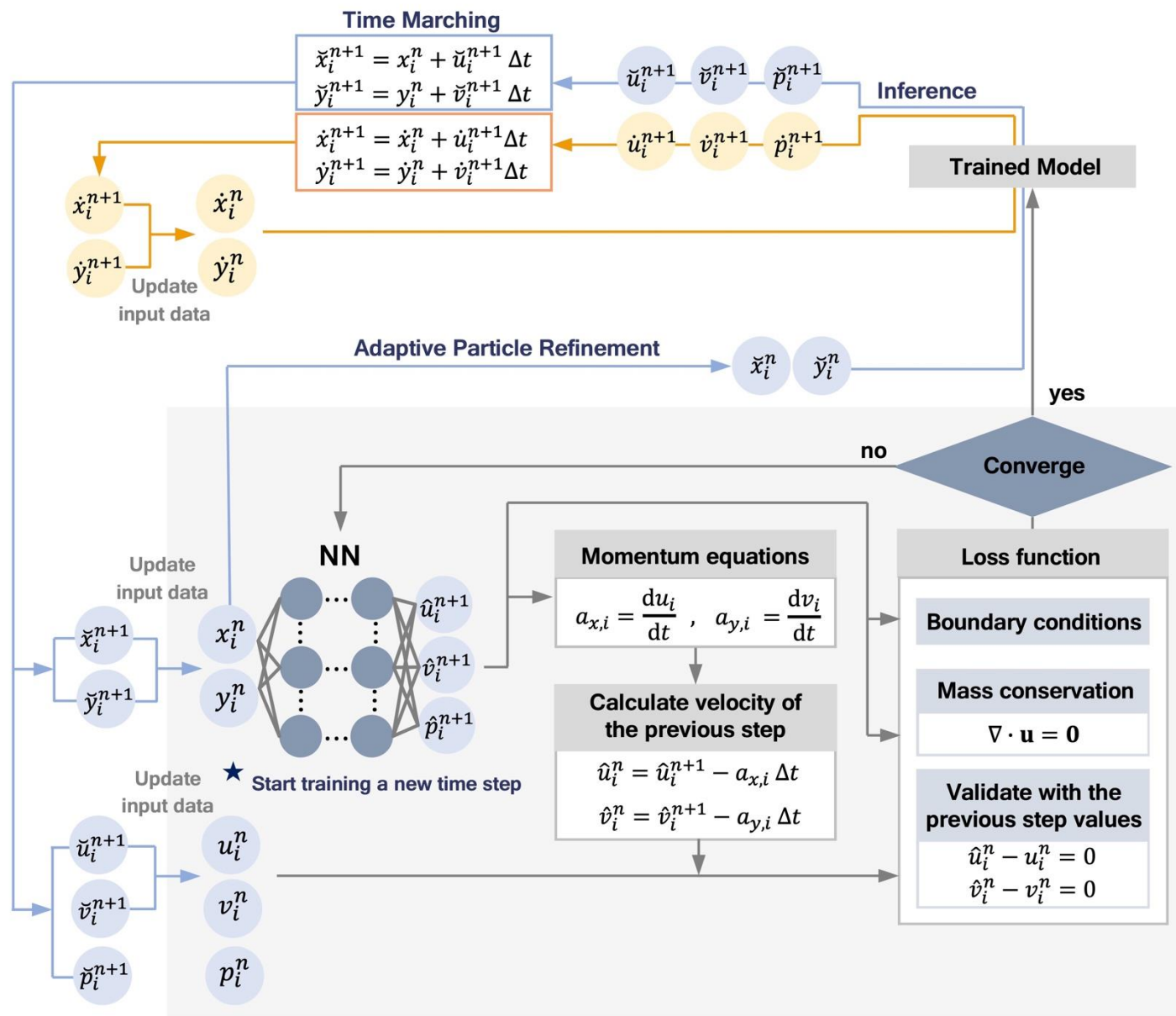
where

$$\mathbf{D} = (\nabla \mathbf{u} + (\nabla \mathbf{u})^T)/2 \quad \|\mathbf{D}\| = \sqrt{\mathbf{D} : \mathbf{D}}$$

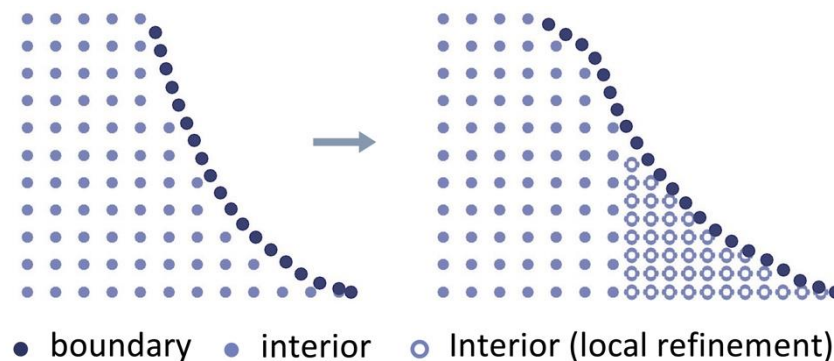
$$\mu(I) = \mu_s + \frac{\mu_2 - \mu_s}{I_0/I + 1} \quad I = \frac{\sqrt{2}d_g \|\mathbf{D}\|}{\sqrt{p/\rho_g}}$$

$$\mathbf{T} = \frac{\sqrt{2}\mu(I)p}{\|\mathbf{D}\| + \varepsilon} \mathbf{D}$$

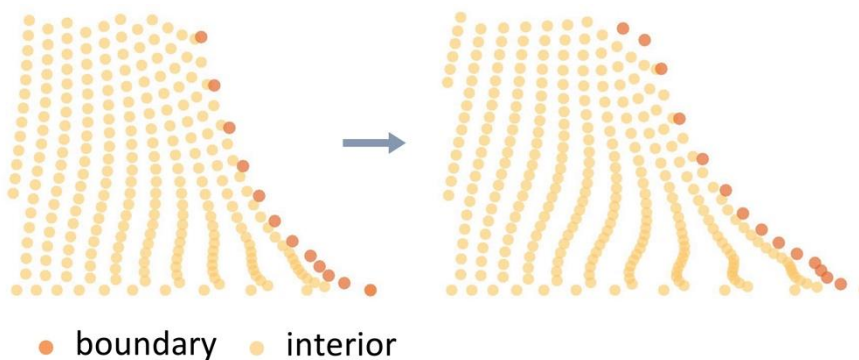
Flowchart of Adaptive Refinement NPM (arNPM)



Training Particles

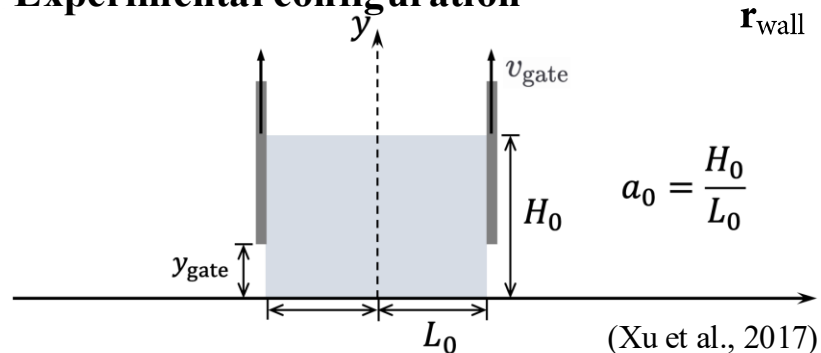


Inference Particles



Test Case: Column Collapse (Glass Beads)

Experimental configuration

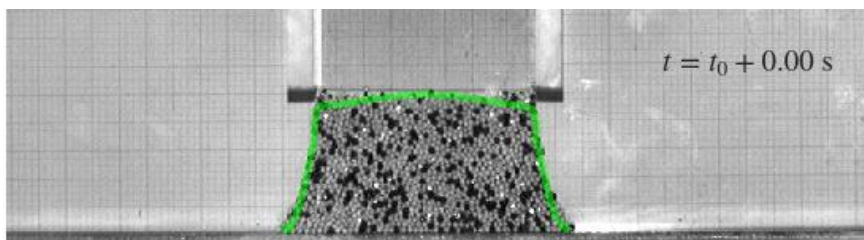


$$\mathbf{r}_{\text{wall}} = -2 \frac{\mu_w p}{W \rho \|\mathbf{u}\|} \frac{\mathbf{u}}{\|\mathbf{u}\|} + \kappa$$

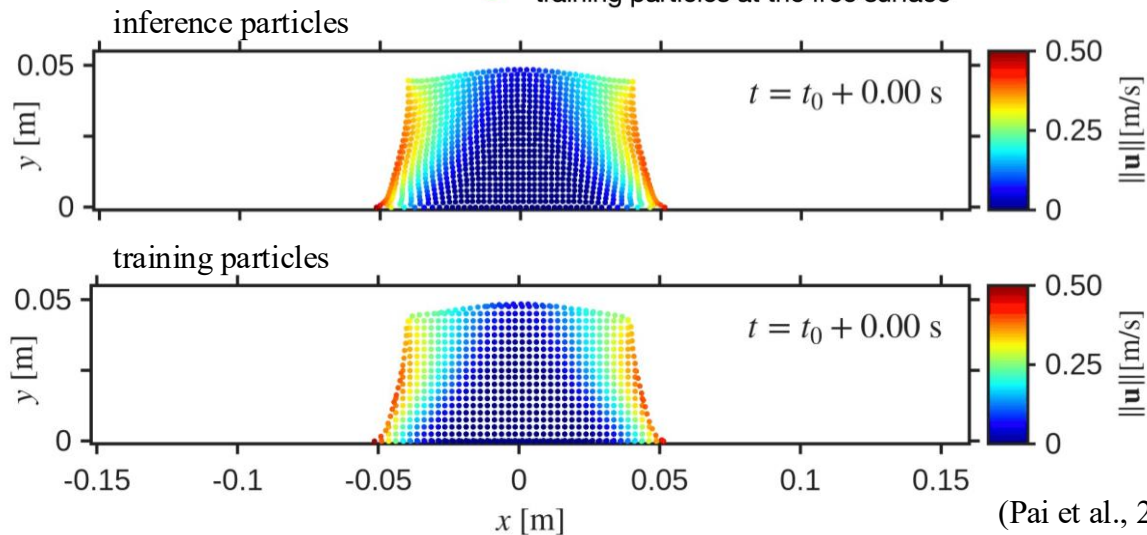
(Roberts, 1969)

Results of arNPM ($a_0 = 1.25$)

$d = 2 \text{ mm}$

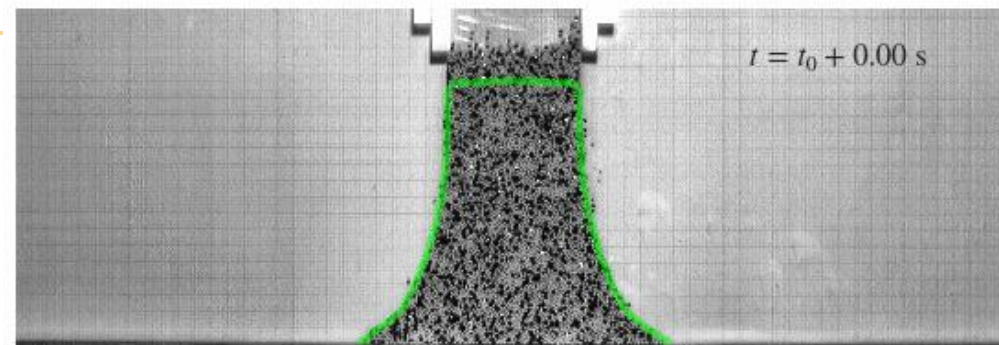


● training particles at the free surface

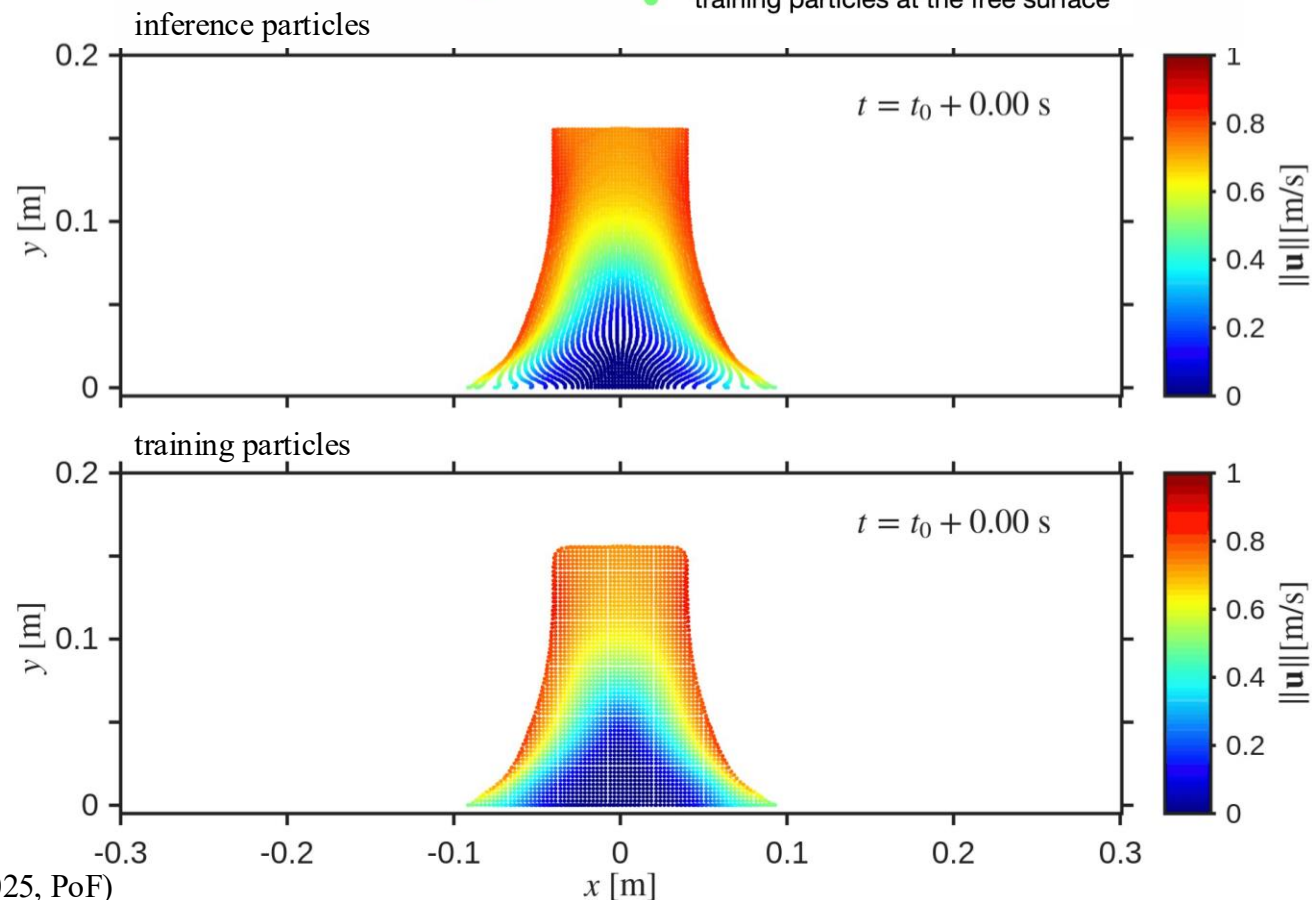


Results of arNPM ($a_0 = 5$)

$d = 2 \text{ mm}$

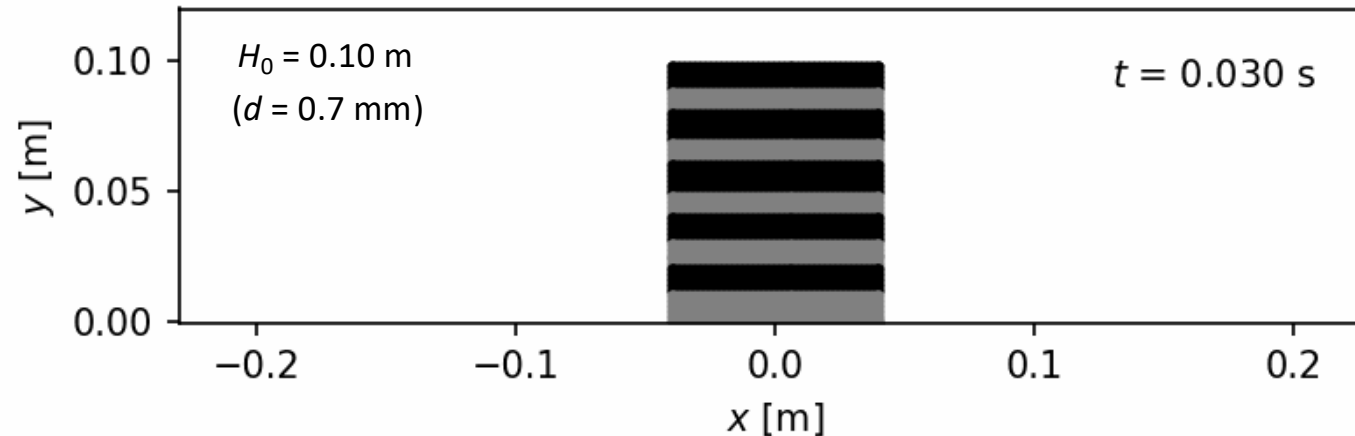
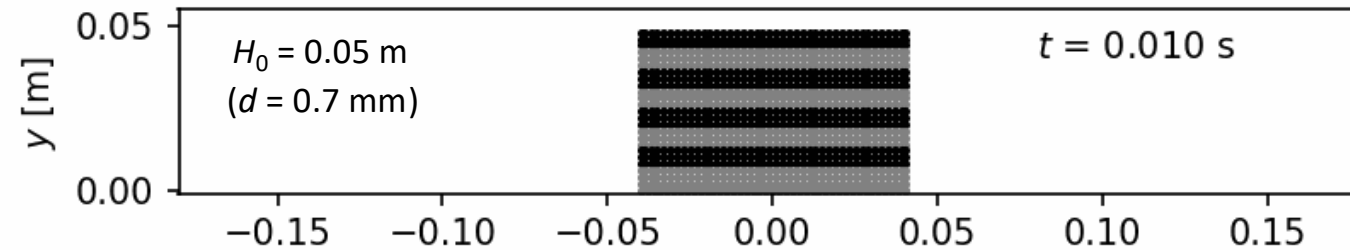
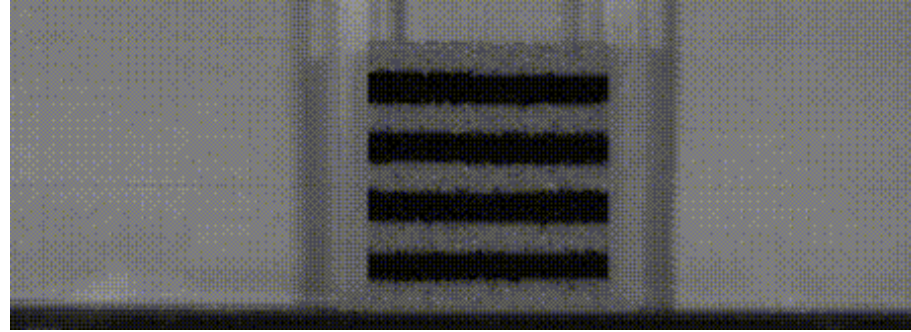


● training particles at the free surface



(Pai et al., 2025, PoF)

Dynamics of Granular Stratification (Ottawa Sand)



Conclusion Remarks

- Bridges physics-informed deep learning approach with granular flow modeling
- Inference particles enable realistic depiction of flow paths
- Provides insights into solid-like cores and fluid-like outer layers
- Agreement with experimental images confirms model validity
- Offers an alternative to SPH/MPS for high-resolution flow prediction

Collaborators

Dr. Yih-Chin TAI (戴義欽博士), Professor
Department of Hydraulic and Ocean Engineering
National Cheng Kung University, Tainan, Taiwan



Dr. Luca SARNO, Assistant Professor
Ms. Pei-Hsin PAI (白佩鑫), Ph.D. Student
Department of Civil Engineering
University of Salerno, Salerno, Italy

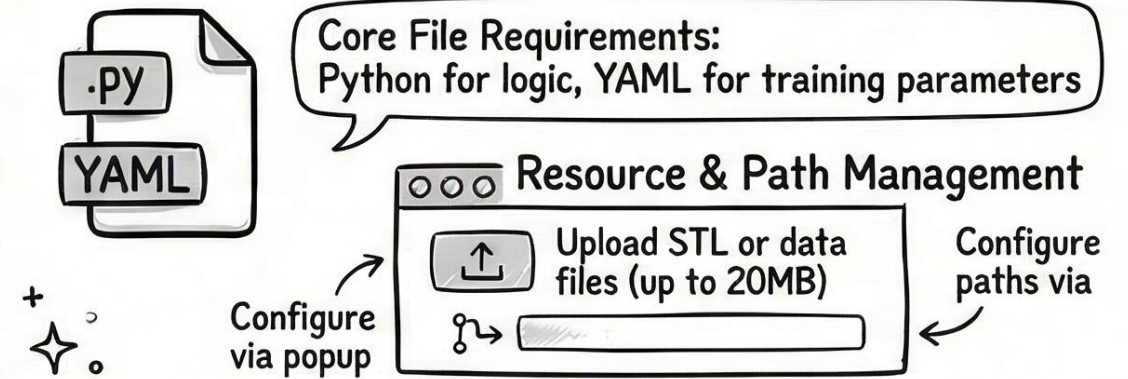


Master the NCHC PINNs Computing Platform

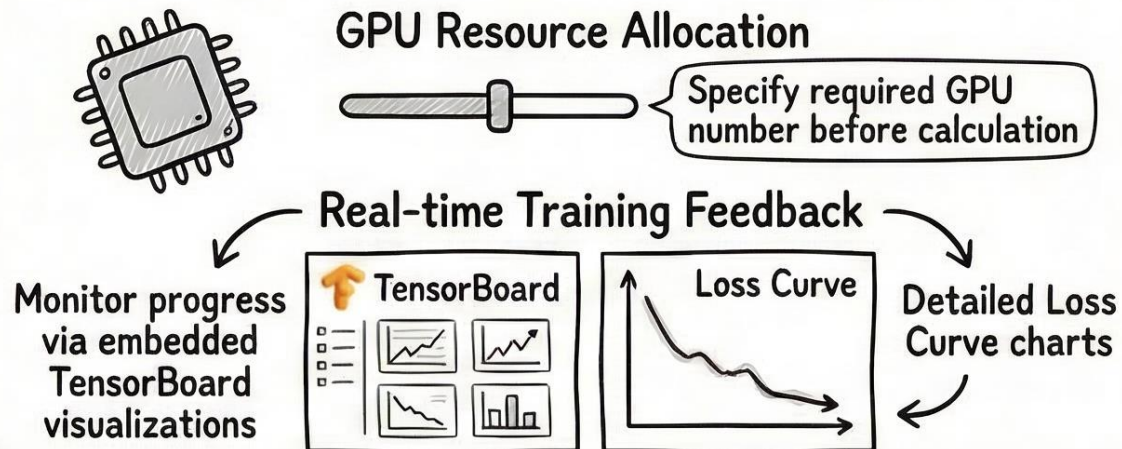
1. Secure Entry & System Access



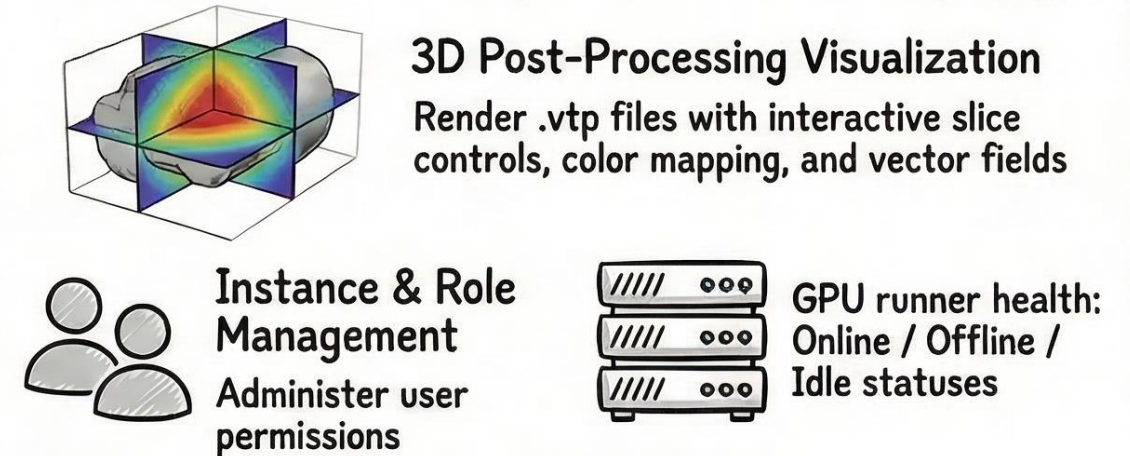
2. Project Initiation & Configuration



3. Computing & Live Monitoring



4. Advanced Analysis & Admin



Just PINNs it!

